Lab 2

DSA

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Exercise

A weather station records temperature readings from multiple sensors. Each sensor has a unique ID and a temperature value.

Write a C program to perform the following tasks [CO1] :

- Define a structure to store sensor information.
- Dynamically allocate memory for `n` sensors using `malloc`.
- Read sensor data using pointer-based traversal.
- Traverse the data using pointers to find the maximum temperature.
- Display all sensor readings and the maximum temperature.

Instructions

1. **Part 1:** Write the logic (set of steps) to solve the problem.
2. **Part 2:** Write a C program based on the logic given in Part 1.

Sample Input

```
Enter number of sensors: 3

Enter data for sensor 1
Sensor ID: 101
Temperature: 32.5

Enter data for sensor 2
Sensor ID: 102
Temperature: 29.8

Enter data for sensor 3
Sensor ID: 103
Temperature: 35.2
```

Sample Output

```
Sensor Readings:
Sensor ID: 101, Temperature: 32.50
Sensor ID: 102, Temperature: 29.80
Sensor ID: 103, Temperature: 35.20

Maximum Temperature: 35.20
```

Assessment Rubric

Assessment Criteria	Marks
Part 1: Algorithm / Logic (Set of Steps) Clear and correct logical steps covering structure definition, memory allocation, input, traversal, maximum temperature computation, and memory deallocation	2
Structure Definition and Usage Correct definition of structure with appropriate data members (sensor ID and temperature) and proper usage in the program	2
Dynamic Memory Allocation Correct use of `malloc`, proper size calculation, and NULL pointer check	2
Pointer-Based Traversal and Data Processing Correct reading of data using pointers, pointer traversal, and accurate computation of maximum temperature	3
Output Formatting and Memory Deallocation Proper display of sensor readings and maximum temperature, and correct use of `free()`	1